

What are the advantages and limitations of using script-based parametric modelling (e.g., OpenSCAD), compared to graphical parametric tools, in the design of personalised upper limb prosthetics, particularly in supporting Design for Additive Manufacturing and user-centred design outcomes? I am developing a master’s dissertation in Industrial Design focused on parametric modelling for personalised upper limb prosthetics within a Research Through Design framework. Focus on: – Technical structuring of parameters, constraints, and dependencies – Integration of anthropometric data into parametric models – Strategies for ensuring printability and structural integrity – Advantages such as reproducibility, scalability, and automation – Limitations related to usability, accessibility, and complexity

No study in this literature directly compared script-based and graphical parametric modeling for upper limb prosthetics, but the repeated convergence of graphical-tool users toward externalized, script-like automation strategies—alongside persistent gaps in embedded DfAM constraints and clinical accessibility—strongly suggests that script-based tools offer untapped advantages in reproducibility, conditional geometry logic, and manufacturing rule integration, with their principal limitation being a usability barrier for non-engineering clinical users.

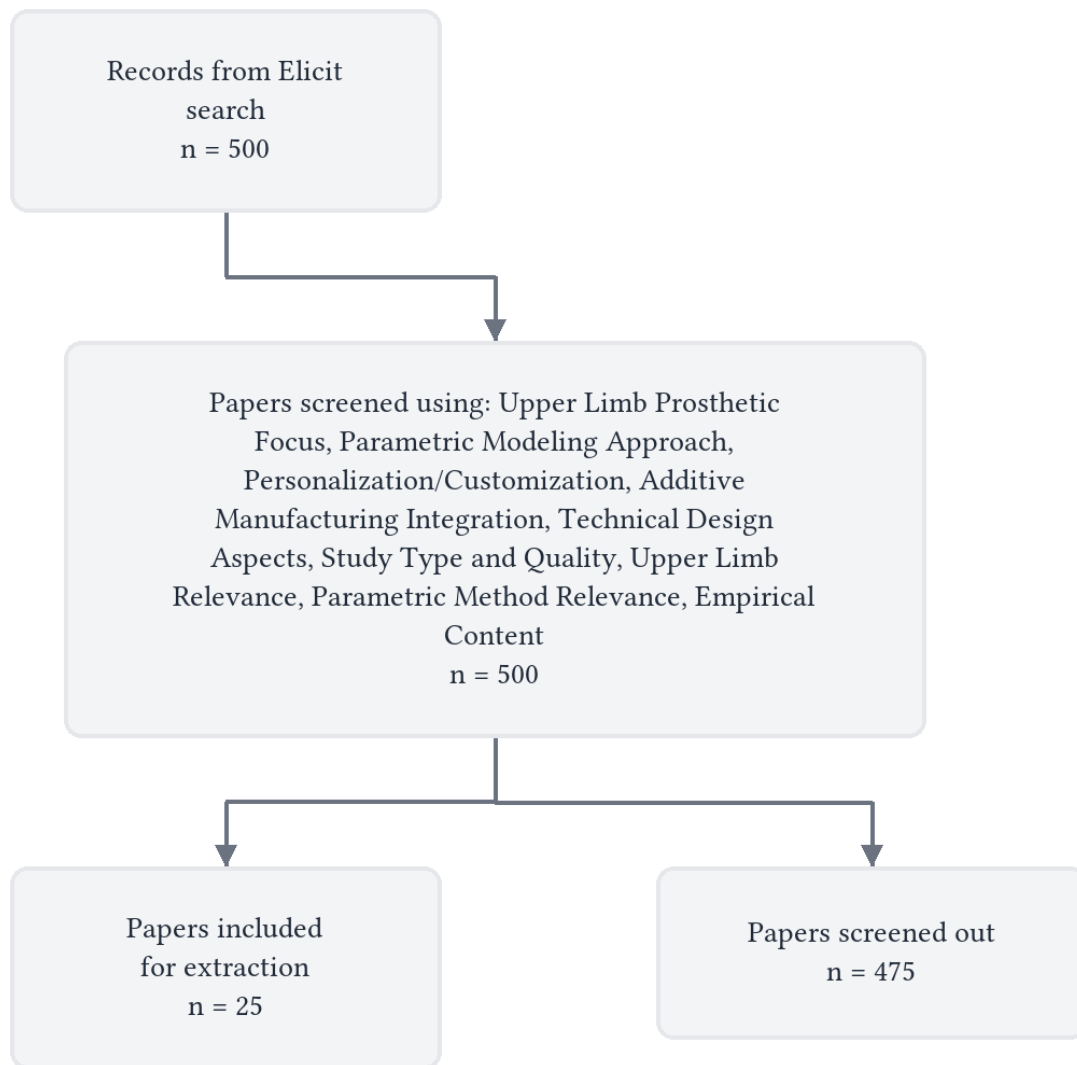
Abstract

No included study employed a script-based parametric modeling tool such as OpenSCAD for upper limb prosthetic design; all 25 sources that specified their software used graphical interface-based CAD platforms—primarily SolidWorks, Autodesk Fusion 360, and Autodesk Inventor [1–4]—sometimes augmented with external automation via Excel spreadsheets [4, 5], iLogic rule engines [3], or MATLAB scripting [6]. Direct comparative evidence between script-based and graphical parametric approaches therefore does not exist in this literature. However, recurring workflow patterns are informative: multiple studies identified CAD expertise as a bottleneck [4, 5] and converged on externalizing parameter control through configuration files, spreadsheets, or embedded scripting [2, 4, 5, 7]—strategies that functionally replicate core features of script-based tools. Demonstrated advantages of parametric modeling include superior customization fidelity over uniform scaling [1, 8], rapid design turnaround (15–45 minutes from scan to print-ready model in automated pipelines) [4], significant cost reduction (as low as USD 50) [3], and accessibility for non-CAD specialists when appropriate interfaces are provided [9, 10]. Design for Additive Manufacturing constraints—such as print orientation, tolerances, and wall thickness—were inconsistently integrated into parametric models, with only two studies explicitly optimizing print orientation [3, 6], suggesting an area where script-based tools could embed manufacturing rules as first-class parameters.

User-centered validation remains limited across the field, with most studies relying on single-user case studies or informal feedback rather than validated instruments [11, 12]; the most rigorous evaluation tested nine amputees using the BAM-ULA protocol and achieved a mean score of 7.4/10 [13]. The functional requirements emerging from this literature—externalized parameter input, conditional geometry adaptation for multiple amputation types [3], automated scan-to-print pipelines [6], and embedded DfAM constraints—are natively supported by script-based parametric tools. The principal barrier to their adoption is likely the usability gap for clinical users, a challenge that graphical-tool studies themselves encountered and addressed through hybrid externalization strategies [2, 4, 5]. The

absence of any script-based implementations in the current evidence base represents a significant gap that precludes definitive comparative conclusions but indicates an unexplored design space with strong theoretical alignment to the automation and reproducibility demands of personalized prosthetic fabrication.

Flow Diagram



Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: "What are the advantages and limitations of using script-based parametric modelling (e.g., OpenSCAD), compared to graphical parametric tools, in the design of personalised upper limb prosthetics, particularly in supporting Design for Additive Manufacturing and user-centred design outcomes?"

I am developing a master's dissertation in Industrial Design focused on parametric modelling for personalised upper limb prosthetics within a Research Through Design framework.

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The search returned 500 total results from Elicit.

We retrieved 500 papers most relevant to the query for screening.

Screening

We screened in sources based on their abstracts that met these criteria:

- **Upper Limb Prosthetic Focus:** Does this study focus on upper limb prosthetic design or development?
- **Parametric Modeling Approach:** Does this research involve parametric modeling approaches (script-based, graphical, or comparative methods)?
- **Personalization/Customization:** Does this study address personalization, customization, or user-specific design considerations?
- **Additive Manufacturing Integration:** Does this research examine additive manufacturing/3D printing integration in prosthetic design?
- **Technical Design Aspects:** Does this study evaluate technical aspects such as parameter structuring, constraints, dependencies, or anthropometric data integration?
- **Study Type and Quality:** Is this study an empirical study, case study, comparative analysis, technical evaluation, systematic review, or meta-analysis (rather than a conference abstract, editorial, or opinion piece without empirical data)?
- **Upper Limb Relevance:** Does this study include upper limb prosthetic applications (rather than focusing exclusively on lower limb prosthetics)?
- **Parametric Method Relevance:** Does this research involve parametric design approaches or provide comparison to parametric methods (rather than focusing solely on non-parametric approaches)?
- **Empirical Content:** Does this study contain empirical data or technical analysis (rather than being only a conference abstract, editorial, or opinion piece)?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Parametric Tool Type:**

Extract the specific parametric modeling tool(s) used for upper limb prosthetic design, including:

- Name and type of software (e.g., OpenSCAD, Grasshopper, Fusion 360, SolidWorks)
- Whether it's script-based (code-driven) or graphical interface-based
- If multiple tools were used or compared
- Any hybrid approaches combining script and graphical methods
- Version information if specified

• **Parameter Architecture:**

Extract details about how the parametric model was technically structured for upper limb prosthetic customization, including:

- Key parameters defined (dimensional, anthropometric, functional)
- Constraint systems implemented
- Dependencies between parameters
- Hierarchical relationships in the parameter structure
- Methods for parameter input/modification
- Any automation or scripting of parameter relationships

• **Anthropometric Integration:**

Extract information about how anthropometric data was incorporated into the parametric prosthetic design, including:

- Source of anthropometric measurements (3D scanning, manual measurement, databases)
- Specific body measurements used (limb dimensions, joint locations, proportional relationships)
- Methods for translating measurements into design parameters
- Accommodation of anthropometric variability
- Any standardization or scaling approaches used
- Integration challenges or solutions mentioned

• **DfAM Implementation:**

Extract strategies used to ensure Design for Additive Manufacturing compatibility and structural integrity in the parametric prosthetic models, including:

- Print orientation considerations built into the model
- Support structure minimization strategies
- Wall thickness and structural reinforcement approaches
- Tolerance and clearance specifications for moving parts
- Material property considerations in the design
- Post-processing requirements integrated into design
- Any printability validation methods used

• **Demonstrated Advantages:**

Extract specific advantages of the parametric modeling approach demonstrated in upper limb prosthetic design, including:

- Reproducibility benefits (consistency across designs, version control)
- Scalability advantages (adaptation to different users, batch customization)
- Automation capabilities (reduced manual design time, systematic variations)

- Customization speed and efficiency gains
- Design iteration and modification ease
- Quality improvements over non-parametric approaches
- Cost or time savings quantified
- Any comparative advantages if multiple approaches were tested

- **Identified Limitations:**

Extract limitations and challenges encountered with the parametric modeling approach for upper limb prosthetic design, including:

- Usability issues (learning curve, interface complexity, user expertise requirements)
- Accessibility barriers (software costs, technical knowledge needs, hardware requirements)
- Design complexity limitations (geometric constraints, feature limitations)
- Integration challenges with existing workflows
- Customization boundaries or restrictions
- Technical performance limitations
- User acceptance or adoption barriers identified
- Any workarounds or mitigation strategies mentioned

- **User-Centered Outcomes:**

Extract outcomes related to user-centered design and actual prosthetic performance for upper limb applications, including:

- User satisfaction measures or feedback
- Functional performance metrics (range of motion, grip strength, task completion)
- Comfort and fit assessments
- Aesthetic acceptance ratings
- Usage patterns or abandonment rates
- User involvement in the design process
- Customization effectiveness for individual needs
- Quality of life impacts measured
- Any user testing or validation results

- **Validation Methods:**

Extract methods used to validate the parametric prosthetic designs for upper limb applications, including:

- Structural analysis techniques (FEA, stress testing, fatigue analysis)
- Functional testing approaches (mechanical testing, user trials)
- Manufacturing validation (print quality assessment, dimensional accuracy)
- Biomechanical validation methods
- User testing protocols
- Comparison studies with existing prosthetics
- Standards or guidelines followed
- Performance metrics and acceptance criteria used

- **Comparative Findings:**

Extract any direct or implicit comparisons between script-based and graphical parametric modeling approaches for upper limb prosthetic design, including:

- Performance differences in design outcomes
- Efficiency comparisons (time, effort, resources)
- Learning curve and usability differences
- Customization capability comparisons
- Integration ease with additive manufacturing workflows
- User preference data between approaches
- Technical capability differences
- Cost-benefit trade-offs identified
- Recommendations for when to use each approach

Results

Characteristics of Included Studies

The 25 sources reviewed encompass a range of study types addressing parametric modeling for upper limb prosthetic design, including primary design studies, case studies, design methodology papers, and material selection analyses. The table below summarizes key characteristics of each included source.

Study	Full text retrieved?	Study Focus	Prosthetic Component	Target Population	Parametric Tool (if specified)
Daniel Lim et al., 2018	Yes	Parametric finger design vs. uniform scaling [1]	Finger [1]	Adults aged 18–30 [1]	Autodesk Fusion 360 [1]
M. Moreo et al., 2016	No	Parametric hand for children in developing countries [2]	Full hand [2]	Children in developing countries [2]	SolidWorks [2]
Daniel Lim et al., 2018a	No	Parametric finger design using four-bar linkage [8]	Finger [8]	Adults aged 18–30 [8]	Not specified [8]
Rodrigo Cézar da Silveira Romero et al., 2025	Yes	Parametric prostheses for multiple amputation levels [3]	Hand (total/partial) [3]	Users with various amputation types [3]	Autodesk Inventor (iLogic) [3]
Alejandro M. Saldarriaga et al., 2024	No	Parametric socket design for transradial prostheses [14]	Socket and forearm [14]	Transradial amputees [14]	Not specified [14]
Julio Fajardo et al., 2020	No	Modular myoelectric prosthesis design [15]	Full hand (myoelectric) [15]	Transradial amputees [15]	Not specified [15]

Study	Full text retrieved?	Study Focus	Prosthetic Component	Target Population	Parametric Tool (if specified)
Ludovica Nini et al., 2024	No	Parametric finger with linkage mechanism and phalangeal shells [16]	Finger [16]	Adult volunteer [16]	Not specified (parametric CAD) [16]
Marlene Bustamante et al., 2018	No	Parametric body-powered hand with Excel-driven customization [5]	Full hand [5]	General upper limb amputees [5]	Excel spreadsheet [5]
Filip Górski et al., 2022	Yes	Automated design via 3D scanning and intelligent CAD models [4]	Mechanical and cosmetic prostheses [4]	Real patients (adults and children) [4]	Autodesk Inventor with iLogic [4]
K. Ráž et al., 2024	Yes	Parametric outer socket production using MJF [17]	Forearm socket [17]	Forearm amputees [17]	Not specified (parametric CAD) [17]
Mixuan Li et al., 2025	No	Parametric design tool for OTs to create hand grips [9]	Assistive hand grips [9]	Individuals with hand impairments [9]	Not specified [9]
J. Lopez et al., 2021	No	Non-assembly 3D printed hand with smartphone-based socket fitting [18]	Full hand and socket [18]	Low-income settings, developing countries [18]	Not specified [18]
Julio Fajardo et al., 2017	No	Open-source multifunctional myoelectric prosthesis [19]	Full hand (myoelectric) [19]	Transradial amputees [19]	Not specified [19]
Junghun Lee et al., 2023	No	Computer vision-based prosthetic customization [20]	Full hand and socket [20]	Resource-constrained environments [20]	Not specified (open-source software) [20]
Enzo Romero et al., 2025	No	AI-driven personalized myoelectric prosthesis [13]	Full hand (myoelectric) [13]	Transradial amputees in low-income regions [13]	Not specified [13]

Study	Full text retrieved?	Study Focus	Prosthetic Component	Target Population	Parametric Tool (if specified)
Kelvin Loutan et al., 2023	No	Low-cost voluntary closing finger with parametric optimization [11]	Finger [11]	Individual amputee [11]	Not specified [11]
C. Chu et al., 2019	No	3D parametric hand model for short thumb orthosis [21]	Thumb orthosis [21]	Taiwanese adults (120 participants) [21]	Not specified [21]
Augustin Lerebours et al., 2019	No	Voxel-based parametric biomechanical model for AM [10]	Trapezium replacement prosthesis [10]	Patients requiring bone replacement [10]	Not specified (voxel-based method) [10]
Yair Herbst et al., 2021	Yes	Fully automated scan-driven pipeline for personalized hand [6]	Full hand [6]	General upper limb amputees [6]	MATLAB and SolidWorks [6]
A. Mohammadi et al., 2018	No	Soft prosthetic hand with parameterized CAD [22]	Full hand (soft robotic) [22]	Upper limb amputees [22]	Not specified (parameterized CAD) [22]
A U Iwuoha et al., 2020	No	3D printed body-powered limb for child amputee [12]	Full hand [12]	12-year-old amputee [12]	Fusion 360 [12]
Wrist Splints et al., 2014	No	Comparison of AM processes for wrist splints [23]	Wrist splint [23]	NHS patients (UK) [23]	Not specified [23]
Ipsita Dhar et al., 2026	Yes	MCDM-based material selection for prosthetic palm [24]	Prosthetic palm [24]	Upper limb amputees [24]	SolidWorks [24]
C. H. Chua et al., 2019	No	Virtual evaluation of prosthetic hand performance [25]	Full hand (multiple designs evaluated) [25]	General upper limb amputees [25]	Not specified [25]

Study	Full text retrieved?	Study Focus	Prosthetic Component	Target Population	Parametric Tool (if specified)
M. Moreo et al., 2016a	No	Parametric hand prosthesis for children (duplicate/variant) [7]	Full hand [7]	Children in developing countries [7]	SolidWorks [7]

Of the 25 sources, six had full texts retrieved (Daniel Lim et al., 2018; Rodrigo César da Silveira Romero et al., 2025; Filip Górski et al., 2022; K. Ráž et al., 2024; Yair Herbst et al., 2021; Ipsita Dhar et al., 2026). The remaining 19 were analyzed based on abstracts only. The studies span publication years from 2014 to 2026, with the majority published between 2018 and 2025. Target populations range from children in developing countries to adults in clinical settings, with a notable emphasis on low-income and resource-constrained contexts. The prosthetic components addressed include individual fingers, full hands, sockets, orthoses, and wrist splints.

A critical observation is that none of the 25 studies employed a script-based parametric modeling tool such as OpenSCAD. All studies that specified their software used graphical interface-based CAD platforms—primarily SolidWorks [2, 7, 24], Autodesk Fusion 360 [1, 12], and Autodesk Inventor [3, 4]—sometimes augmented with automation via external spreadsheets [4, 5] or rule-based scripting within the CAD environment (e.g., iLogic in Autodesk Inventor) [3, 4]. One study used MATLAB alongside SolidWorks for pipeline automation [6], and another proposed a voxel-based approach designed to be accessible to non-CAD specialists [10]. Consequently, direct comparative findings between script-based and graphical parametric approaches are absent from this literature. The analysis below instead characterizes the parametric design landscape as implemented through graphical tools and identifies properties that may inform a comparison with script-based alternatives.

Parametric Tool Selection and Implementation

Study	CAD Platform	Automation Method	Parameter Input Interface	Hybrid Approach
Daniel Lim et al., 2018	Autodesk Fusion 360 [1]	Manual parameter adjustment [1]	GUI-based [1]	No [1]
M. Moreo et al., 2016	SolidWorks [2]	External file linkage [2]	External file (no CAD opening required) [2]	Yes (external file) [2]
Rodrigo César da Silveira Romero et al., 2025	Autodesk Inventor [3]	iLogic rules [3]	CAD-integrated automation [3]	Yes (iLogic scripting within GUI) [3]
Marlene Bustamante et al., 2018	Not specified [5]	Excel-driven dimensions [5]	Excel spreadsheet [5]	Yes (Excel + CAD) [5]
Filip Górski et al., 2022	Autodesk Inventor [4]	iLogic rules + MeshLab [4]	Excel spreadsheets filled semi-automatically [4]	Yes (Excel + iLogic + MeshLab) [4]

Study	CAD Platform	Automation Method	Parameter Input Interface	Hybrid Approach
K. Ráž et al., 2024	Not specified [17]	Algorithmic model from 3D scan [17]	User-defined parameters for thickness and position [17]	Implied (algorithmic generation) [17]
Yair Herbst et al., 2021	MATLAB + SolidWorks [6]	Fully automated pipeline [6]	MATLAB GUI [6]	Yes (MATLAB scripting + SolidWorks CAD) [6]
Ipsita Dhar et al., 2026	SolidWorks [24]	MCDM-based material selection [24]	SolidWorks simulation [24]	No [24]
M. Moreo et al., 2016a	SolidWorks [7]	External file linkage [7]	External file [7]	Yes (external file + CAD) [7]

Among the studies that provided sufficient detail on their parametric workflow, a spectrum of automation emerges. At one end, designs rely on manual parameter modification within a graphical CAD interface [1]. At the other, fully automated pipelines accept scan data and produce print-ready files with minimal human intervention [6]. Several studies occupy an intermediate position by coupling a graphical CAD platform with external data sources—most commonly Excel spreadsheets that allow non-experts to modify design dimensions without opening or understanding the CAD environment [2, 4, 5, 7]. The use of iLogic rules within Autodesk Inventor represents an embedded scripting approach that automates parameter dependencies while remaining within a graphical tool [3, 4]. These hybrid strategies share a functional resemblance to script-based parametric modeling insofar as they externalize parameter control, enable batch customization, and reduce the need for CAD expertise—goals that script-based tools like OpenSCAD achieve natively.

Parameter Architecture and Anthropometric Integration

A central challenge in parametric prosthetic design is defining a parameter set that captures the relevant anthropometric variability while remaining tractable for automation. The studies reveal two broad strategies.

Reduced parameter sets driven by correlation analysis. Moreo et al. (2016/2016a) performed a statistical analysis on eight anthropometric measurements and determined that only four—grip circumference, palm breadth, thumb breadth, and palm length—were independently important, and that all could ultimately be connected to a single master parameter (palm breadth) [2, 7]. This hierarchical reduction simplifies the parametric model considerably but may sacrifice fit accuracy for individuals whose proportions deviate from the population mean. Similarly, Daniel Lim et al. (2018) defined just two independent parameters—finger length (L) and width (W)—and showed that these exhibit negligible correlation in their sample of 50 adults, necessitating their independent specification rather than derivation from a single variable [1, 1].

Scan-driven multi-parameter models. Several studies used 3D scanning to capture patient-specific geometry directly. Górski et al. (2022) fed 3D scan point clouds from patients' stumps and healthy limbs into intelligent CAD models via semi-automatic Excel processing [4]. Ráž et al. (2024) used a prosthetist-made soft socket as the scanning input, from which an algorithmic model generated a parametric outer socket [17]. Herbst et al. (2021) employed RealSense cameras to capture user anatomy and fit piecewise Bezier parametric surfaces to the resulting point cloud

[6]. Chu et al. (2019) collected 19 anthropometric parameters from 120 Taiwanese adults using a scanning device and synthesized individualized 3D hand mesh models [21]. Nini et al. (2024) used a 3D scanner to acquire phalangeal shapes and integrated these into a parametric CAD system for finger shell design [16]. Lopez et al. (2021) attempted automatic anthropometry from smartphone photographs using a Statistical Shape Model, though the resulting error was too large for acceptable socket design, necessitating a fallback to manual tape measurement [18].

The methods for translating measurements into design parameters also varied. Some studies used proportional relationships where component dimensions are set as percentages of a master dimension [1, 2], while others employed constraint-based rules (e.g., iLogic) that manage dependencies between parameters and enforce printability constraints simultaneously [3, 4]. Romero et al. (2025) used a geometric suppression method within iLogic to adapt a single parametric model to four different amputation types (transmetacarpal, partial radial, partial ulnar, and total) [3].

Design for Additive Manufacturing (DfAM) Implementation

Study	Print Orientation	Support Minimization	Tolerance/Clearance	Material Characterization	Structural Validation	Post-Processing
Daniel Lim et al., 2018	Not addressed [1]	Supports generated for overhangs [1]	5 mm standardized interfacing parts [1]	PLA [1]	Not performed [1]	Assembly with 4-40 screws [1]
Rodrigo César da Silva Romero et al., 2025	YXZ plane for strength and accuracy [3]	Implied by accuracy focus [3]	Implied via FEA [3]	ASTM D638 characterization [3]	FEA under static forces [3]	Not detailed [3]
Filip Górski et al., 2022	Not detailed [4]	Not detailed [4]	Not detailed [4]	Strength values at break recorded [4]	Bending tests [4]	Chemical/mechanical surface treatment needed [4]
K. Ráž et al., 2024	Not detailed [17]	Not detailed [17]	Max 1 mm surface deviation [17]	PA12GB characterized (tensile strength, elongation) [17]	Surface correspondence analysis [17]	Arbitrary surface treatment possible [17]
Yair Herbst et al., 2021	Optimized for strength and minimal support [6]	Optimized layout [6]	Simplified assembly implies consideration [6]	ABS or PLA [6]	50K cycle mechanical testing [6]	Not detailed [6]

Study	Print Orientation	Support Minimization	Tolerance/Clearance	Material Characterization	Structural Validation	Post-Processing
Mixuan Li et al., 2025	Not detailed [9]	Not detailed [9]	Not detailed [9]	Temperature, layer height, infill, material investigated [9]	Not detailed [9]	Not detailed [9]
Augustin Lerebours et al., 2019	Not detailed [10]	Not detailed [10]	Not detailed [10]	Not detailed [10]	Not detailed [10]	SLM post-treatments noted [10]
Wrist Splints et al., 2014	Not detailed [23]	Not detailed [23]	Not detailed [23]	Multi-material (hinges, cushions) [23]	Not detailed [23]	Not detailed [23]
Ipsita Dhar et al., 2026	Not detailed [24]	Not detailed [24]	Not detailed [24]	Mechanical and physical properties evaluated via MCDM [24]	Static analysis in SolidWorks [24]	Not detailed [24]
M. Moreo et al., 2016/2016a	Not detailed [7]	Not detailed [7]	Not detailed [7]	Not detailed [2]	Pinch force testing (35 N achieved) [2]	3-part print, glue and elastic assembly [7]

DfAM considerations were inconsistently addressed across the literature. Only two studies explicitly optimized print orientation: Romero et al. (2025) selected the YXZ printing plane for superior mechanical strength and dimensional accuracy [3], and Herbst et al. (2021) optimized part orientation to maximize strength while minimizing support volume [6]. Tolerance specifications were rarely quantified; Ráž et al. (2024) specified a maximum allowable surface deviation of 1 mm between the scanned input and the generated parametric socket [17], while Lim et al. (2018) used standardized 5 mm interfacing dimensions for assembled components [1].

Material characterization was performed rigorously in only a few studies. Romero et al. (2025) followed ASTM D638 standards for tensile testing of printing material [3], and Dhar et al. (2026) applied multi-criteria decision-making methods (LOPCOW, COBRA, EDAS) to systematically evaluate 11 candidate 3D printing materials across eight mechanical, physical, and economic criteria, identifying acrylic as the optimal choice [24]. Most studies simply specified PLA [1] or ABS [6] without formal characterization. Structural validation via finite element analysis (FEA) was conducted by Romero et al. (2025) [3] and Nini et al. (2024) [16], while Herbst et al. (2021) performed 50,000-cycle mechanical endurance tests on actuated components [6].

The general underspecification of DfAM parameters in the parametric models is notable. Most studies treated the parametric design and the manufacturing process as sequential rather than integrated concerns—a pattern that may partly reflect the limitations of graphical CAD tools in encoding manufacturing constraints as first-class parameters within the design model.

Demonstrated Advantages of Parametric Approaches

The advantages reported across studies cluster into several categories.

Customization fidelity. The most consistently demonstrated advantage is superior fit compared to uniform scaling. Lim et al. (2018) showed that parametrically modeled prosthetic fingers closely match the natural range-of-motion path, while uniformly scaled fingers exhibit poor alignment [1, 8]. This finding is grounded in the empirical observation that finger length and width are not correlated ($r \approx$ negligible across 50 adults) [1], meaning uniform scaling inevitably distorts proportions for most users.

Design speed and efficiency. Górski et al. (2022) reported that their automated pipeline produces a print-ready prosthetic model in 15 to 45 minutes from 3D scan input [4], compared to traditional prosthetic fabrication processes that can take weeks to months [20]. Ráž et al. (2024) noted that their parametric socket workflow replaces time-consuming composite lamination technology and accelerates delivery time [17]. Saldarriaga et al. (2024) described optimization of design time relative to conventional prosthetic procedures [14].

Cost reduction. Romero et al. (2025) estimated a prosthesis cost of approximately BRL 300 (USD 50.15) [3]. Multiple studies emphasized the affordability of 3D-printed parametric prosthetics for low-income settings [13, 15, 18, 20]. Górski et al. (2022) described their output as a fraction of the cost of commercial solutions [4].

Accessibility for non-experts. Several designs specifically aimed to reduce the need for specialist CAD knowledge. Bustamante et al. (2018) allowed customization via an Excel spreadsheet without opening the CAD file [5]. Moreo et al. (2016) similarly linked the SolidWorks model to an external file for non-specialist modification [2, 7]. Lerebours et al. (2019) designed a voxel-based approach explicitly for medical employees unfamiliar with CAD [10]. Li et al. (2025) received positive feedback from occupational therapists on ease of use and real-time design capabilities [9].

Scalability and modularity. The parametric approach enables adaptation to different amputation levels (e.g., Fajardo et al., 2020 designed for varying transradial levels [15]) and different user demographics (e.g., Moreo et al., 2016 for growing children [2]). Romero et al. (2025) accommodated four distinct amputation types within a single parametric framework using geometric suppression [3].

Identified Limitations and Challenges

Limitation Category	Studies Reporting	Key Details
Durability concerns	M. Moreo et al., 2016; M. Moreo et al., 2016a	Primary durability limitation acknowledged; improvement needed in support and printing strategies [2, 7]
Uniform scaling inadequacy	Daniel Lim et al., 2018; Bustamante et al., 2018	Many existing 3D-printed designs only permit uniform scaling, failing to match user-specific anthropometry [1, 5]
CAD expertise requirements	Bustamante et al., 2018; Górski et al., 2022	High customization requires prior CAD knowledge; specialized engineering knowledge needed [4, 5]

Limitation Category	Studies Reporting	Key Details
Measurement accuracy	J. Lopez et al., 2021; Górski et al., 2022	Automatic photogrammetric measurement error too large for acceptable socket design [18]; manual processing introduces inaccuracies [4]
Structural and joint durability	Rodrigo César da Silveira Romero et al., 2025	Mesh interpolation causes localized stress peaks; joints require reinforcement [3]
Fine motor performance	Enzo Romero et al., 2025	Only 22.2% success rate on fine motor tasks despite 100% on gross motor tasks [13]
Aesthetic vs. functional trade-offs	C. H. Chua et al., 2019	Thumb position and digit abduction chosen for aesthetics rather than efficacy; anthropomorphic designs produce unstable grasps [25]
Independent finger control	A U Iwuoha et al., 2020	Inability to control each finger independently; requires additional bionic technologies [12]
Socket design automation	Filip Górski et al., 2022	Difficulty automating socket design; surface modeling complexity in CAD systems [4]
Manufacturing process suitability	Wrist Splints et al., 2014	FDM considered least suitable for upper extremity splinting; LS, SLA, and material jetting require further development [23]
Simulated vs. real-world properties	Ipsita Dhar et al., 2026	Material selection based on simulated properties may not capture real-world complexities such as long-term durability or biocompatibility [24]
High abandonment rates	C. H. Chua et al., 2019; Daniel Lim et al., 2018	56% device abandonment rate cited [1]; design complexity not correlated with improved performance [25]
Temporal adaptability	Kelvin Loutan et al., 2023	Designs may not accommodate changes due to swelling and ageing over time [11]

The limitations reveal a tension between design automation and clinical acceptability. While parametric models excel at generating geometrically customized forms rapidly, challenges persist in ensuring structural durability [2, 3], achieving fine motor dexterity [13], and obtaining accurate input measurements without specialist equipment [4, 18]. The requirement for CAD expertise remains a barrier in graphical tool-based workflows [4, 5], which multiple studies attempted to mitigate through Excel-based interfaces [4, 5] or clinician-facing tools [9].

User-Centered Outcomes and Validation

Study	Validation Method	Participants	Key Functional Outcomes	User Satisfaction
Daniel Lim et al., 2018	ROM simulation comparison [1]	50 adults (anthropometric data) [1]	Parametric model closely matches natural ROM [1]	Not assessed [1]
M. Moreo et al., 2016	Mechanical testing + user trials [2]	Children (age not specified) [2]	Pinch force up to 35 N; functional for daily activities [2]	Not formally assessed [2]
Rodrigo César da Silveira Romero et al., 2025	FEA + tensile testing (ASTM D638) [3]	No user trials conducted [3]	Adaptable gripping for varied object shapes [3]	Not assessed [3]
Filip Górski et al., 2022	3D scan fitting + bending tests [4]	Real patients [4]	Satisfactory visual and mechanical properties [4]	Not formally assessed [4]
K. Ráž et al., 2024	Surface correspondence analysis [17]	Not detailed [17]	Functionality verified; max 1 mm deviation [17]	Increased comfort reported [17]
Mixuan Li et al., 2025	User study with OTs [9]	12 occupational therapists [9]	Not quantified [9]	Positive feedback on ease of use and customization [9]
J. Lopez et al., 2021	Fatigue testing + field testing [18]	Users in Colombia [18]	Bio-inspired design achieved superior pinch force [18]	Field testing conducted [18]
Enzo Romero et al., 2025	BAM-ULA protocol [13]	9 transradial amputees [13]	Mean BAM-ULA 7.4/10 (SD 0.7); 100% gross motor, 22.2% fine motor success [13]	Comparable to high-end commercial devices [13]
Kelvin Loutan et al., 2023	Iterative user testing [11]	1 amputee user [11]	Desirable actuation and grasping achieved [11]	User feedback on fit, comfort, functionality collected [11]
C. Chu et al., 2019	QUEST usability evaluation [21]	1 client (case study) [21]	Greater flexibility and stronger support than traditional orthosis [21]	Mostly high QUEST scores [21]
Yair Herbst et al., 2021	ADL testing + 50K cycle endurance [6]	Prospective users and healthy subjects [6]	Successful ADL performance; ~300 g total weight [6]	Appreciation for low weight and gesture configurations [6]

Study	Validation Method	Participants	Key Functional Outcomes	User Satisfaction
A U Iwuoha et al., 2020	Weight assessment + user trial [12]	1 child amputee (age 12) [12]	188 g weight; comfortable socket; quick state changes [12]	Better functionality than conventional limbs reported [12]
C. H. Chua et al., 2019	Virtual SHAP simulation [25]	No human participants [25]	Clamp-Hand highest score (0.50); Raptor-Hand lowest (0.29) [25]	Not assessed [25]
Ludovica Nini et al., 2024	FEA + deviation analysis + trajectory comparison [16]	1 adult volunteer [16]	Fingertip trajectory compared with human movement [16]	Not assessed [16]

The most rigorous functional evaluation was conducted by Enzo Romero et al. (2025), who tested nine transradial amputees using the BAM-ULA protocol and achieved a mean score of 7.4/10, with a notable disparity between gross motor (100% success) and fine motor (22.2% success) tasks [13]. Chu et al. (2019) applied the validated QUEST instrument and reported mostly high satisfaction scores, with the parametric orthosis outperforming a traditional manually-formed device in flexibility and support [21]. Most other studies relied on informal user feedback or mechanical testing alone, and sample sizes were generally very small—frequently single-user case studies [11, 12, 21]. The virtual evaluation by Chua et al. (2019) is notable for removing user bias and enabling standardized grasp quality comparison across five prosthetic designs, finding that design complexity did not correlate with performance [25, 25].

User involvement in the design process was explicit in Loutan et al. (2023), where the amputee participated in conceptualization and testing phases through two design iterations [11], and in Herbst et al. (2021), where user scan data and preferences drove the automated pipeline [6]. However, the majority of studies did not involve end users in the parametric design workflow itself, instead using anthropometric data as a proxy for personalization.

Synthesis

The central research question asks how script-based parametric modeling (e.g., OpenSCAD) compares to graphical parametric tools in supporting DfAM and user-centered design for upper limb prosthetics. The literature does not provide a direct answer: no included study used a script-based parametric tool, and no study performed a comparative evaluation between the two paradigms. However, the patterns observed across the 25 graphical-tool-based studies illuminate specific functional requirements and workflow bottlenecks that allow for an informed analysis of where each approach might offer advantages or encounter limitations.

The expertise barrier and the push toward externalized parameter control. Multiple studies identified the need for CAD expertise as a significant bottleneck in parametric prosthetic workflows [4, 5]. The recurring solution was to externalize parameter input—through Excel spreadsheets [4, 5], external configuration files [2, 7], MATLAB GUIs [6], or iLogic rule engines [3, 4]. This externalization effectively replicates a core feature of script-based parametric modeling, where the design is defined by a text file of parameters and relationships that can be modified without interacting with a graphical interface. The fact that graphical-tool users converge on script-like solutions for clinical deployment suggests that script-based approaches may offer a more natural workflow for parametric prosthetic

customization, particularly in settings where clinicians or technicians (rather than design engineers) must drive the customization process [9, 10].

Automation and reproducibility. The fully automated pipeline of Herbst et al. (2021)—spanning 3D scanning, parametric surface fitting, print optimization, and PCB layout generation—demonstrates the ceiling of what automation can achieve within a hybrid MATLAB/SolidWorks framework [6]. Script-based tools would natively support this level of automation through programmatic geometry generation. Górski et al. (2022) achieved 15–45 minute turnaround from scan to print-ready model using their iLogic-automated Inventor pipeline [4], establishing a benchmark for design speed that script-based approaches would need to match. The reproducibility advantage of parametric modeling was consistently cited [1, 11, 22], and script-based tools would inherently provide version control through their text-based design files—a feature not available in binary CAD formats without dedicated PLM systems.

DfAM integration gaps. The underspecification of manufacturing constraints within parametric models is a persistent finding. Print orientation was explicitly optimized in only two studies [3, 6], and tolerance specifications for moving parts were rarely formalized [1, 17]. Script-based parametric tools could potentially embed DfAM rules (minimum wall thickness, clearance gaps, overhang angle limits) as explicit constraints within the design script, making them intrinsic to the model rather than afterthoughts. The geometric suppression approach used by Romero et al. (2025) to adapt a single model to four amputation types [3] demonstrates the kind of conditional logic that script-based tools handle natively through if/else constructs, whereas graphical tools require specialized automation add-ons like iLogic.

The complexity-usability trade-off. Graphical tools provide immediate visual feedback and lower initial learning barriers, which may explain their universal adoption in this literature. The positive reception of parametric interfaces by occupational therapists in Li et al. (2025) [9] and the emphasis on accessibility for non-CAD specialists in Lerebours et al. (2019) [10] underscore the importance of usability in clinical contexts. Script-based tools, by contrast, require comfort with code syntax and offer less immediate spatial reasoning support, which may limit adoption among clinical users. However, for engineering-driven design teams producing prosthetics at scale, script-based tools may offer advantages in maintainability, testability, and integration with computational pipelines—advantages that align with the automated workflows several studies aspired to achieve [4, 6, 20].

Anthropometric parameter structure. The hierarchical parameter architectures described—from single-master-parameter models [2] to multi-parameter scan-driven systems [6, 21]—are equally implementable in script-based and graphical environments. The key differentiator is how parameter dependencies are expressed and maintained. In graphical tools, dependencies are often implicit in the feature tree and may break unpredictably when parameters change beyond tested ranges [4]. In script-based tools, dependencies are explicit in code, making them easier to audit, test, and debug—but also requiring the designer to manage geometric construction logic manually rather than relying on direct modeling affordances.

In summary, the literature reveals a field that has converged on graphical parametric CAD tools augmented with various forms of automation scripting, without exploring dedicated script-based alternatives. The functional requirements identified—externalized parameter control, automated pipelines, embedded manufacturing constraints, and conditional geometry adaptation—are well suited to script-based parametric modeling. The principal barrier to script-based adoption is likely the usability gap for clinical and non-engineering users, a barrier that the graphical-tool studies themselves struggled to overcome and addressed through hybrid externalization strategies that partially replicate script-based functionality.

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